Fluid Dynamics and Bernoulli's Principle

SYNOPSIS

Note 1 was fluid standing still. Everything there came from one idea — the fluid above you has weight. Now we set the fluid moving, and two new ideas take over, both of them conservation laws you already know, rewritten for something that flows.

Mass is conserved, so whatever goes into a pipe has to come out of it. Squeeze the pipe and the fluid has no choice but to speed up — that is the **continuity principle**.

Energy is conserved, so a fluid that speeds up has to pay for the extra kinetic energy out of something else. What it pays with is *pressure*. That is **Bernoulli's equation**, and it is the work–energy theorem with the terms divided through by volume.

Put the two together and you get the result that sounds wrong the first time you hear it: **where a fluid moves faster, its pressure is lower.** It is why a wing lifts, why a shower curtain gets sucked inward, and why the narrow part of a pipe is the part at risk of collapsing.

BEFORE YOU READ ON — PREDICT

A pipe carrying water narrows to half its width and then opens out again. In the narrow section, is the water pressure **higher** than in the wide section, **lower**, or the **same**? Commit to one in your guided notes, with a reason, before you read on.

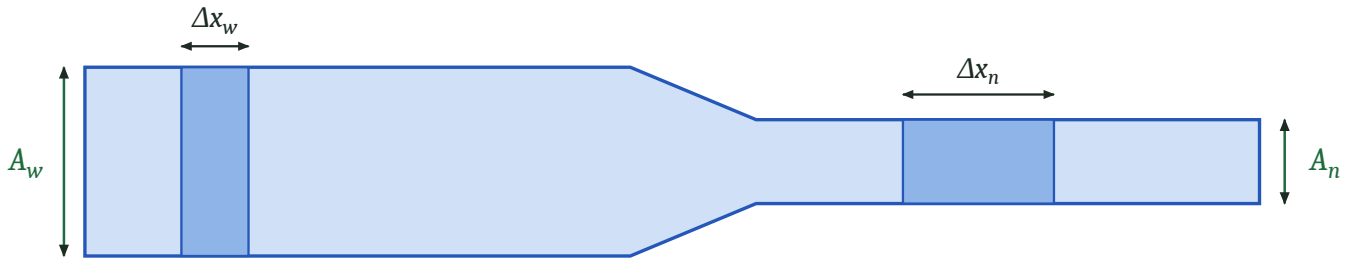
PERFORMANCE EXPECTATIONS

State the continuity principle and use $A_1v_1 = A_2v_2$. Explain where Bernoulli's equation comes from, term by term, and say what each term is the energy of. Use it to compare two points on a streamline. Explain why faster flow means lower pressure, and apply that to a wing.

For PHY 151 — with calculus: Get Bernoulli's equation by integrating Euler's equation along a streamline, see exactly which assumptions the integral needs, and recover Torricelli's result for a draining tank as a special case.

Continuity — the fluid has to go somewhere

Water goes into one end of a pipe and comes out of the other. None of it is created on the way through and none of it disappears, so in any given second, **the amount that enters has to equal the amount that leaves**. That is mass conservation, and for a fluid of constant density it is volume conservation too. The product Av is the **volume flow rate**, Q, in m³/s — litres per second by another name — and continuity says Q is the same everywhere along the pipe. Now watch what that forces on a pipe that changes width.



Drawn as a duct of fixed depth, so the **height on the page is the area**, and the wide section has 2.25× the area of the narrow one. Draw a speed arrow in each section, to lengths that are in the right ratio, and label them. Then write under each section whether it is the fast one or the slow one, and say what is the same about the two shaded slugs.

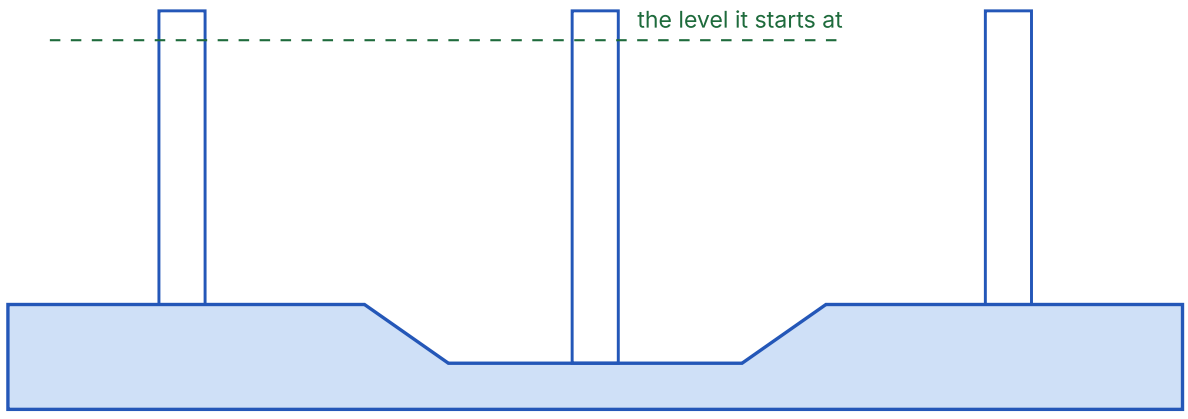
1. In a time Δt the slug in the wide section travels Δx_w and the slug in the narrow section travels Δx_n. (A) Write the statement that the two shaded volumes are equal. (B) Divide it by Δt and say what each ratio has become. (C) Which is larger here, v_w or v_n, and why?
2. Water flows at 1.2 m/s through a pipe of cross-sectional area 0.040 m². (A) Find Q, in m³/s and in litres per second. (B) The pipe narrows to 0.015 m². What is the speed there?

3. You put your thumb over the end of a hose and the water reaches further. Of A, v and Q, say which changed, which did not, and why.

Fast flow is low-pressure flow

Here is where intuition goes wrong. Squeezing a pipe *sounds* like it should raise the pressure inside the squeeze — you are cramming the same water through a smaller hole. The opposite happens. The narrow section is the **low**-pressure section.

The reason is energy. The fluid arrives at the throat moving slowly and leaves it moving fast, so its kinetic energy went up. Nothing pushed it but the fluid behind, so that energy had to come from somewhere, and the only account left to draw on is pressure. The fluid spends pressure to buy speed.



Each standpipe reads the pressure in the duct below it — the taller the column, the higher the pressure. Shade the water column in each of the three, to heights that tell the right story against the dashed

starting level, and label each one high or low. Draw a speed arrow in the wide section and in the throat, and say what the third column tells you that the second cannot.

4. True or false, with one line of reason each. (A) Squeezing the pipe raises the pressure inside the squeeze. (B) The fluid speeds up because something pushed it harder. (C) Energy is lost at the throat.
5. The three standpipes read the pressure beneath them. (A) Which column is the shortest, and why? (B) The third column stands as tall as the first. What does that tell you that the second one on its own does not?
6. Write "narrow → fast → low pressure" out as a chain, and name the conservation law behind each arrow.

Bernoulli's equation — energy conservation for a fluid

Now make that bookkeeping exact. Take the work–energy theorem you used for blocks and carts in Module 4, apply it to a lump of fluid of volume V, and divide every term by V. Gravity does work on the lump, and so does the surrounding fluid, which pushes it along with a force PA over a distance — a work of $P \times$ volume. Nothing new goes in; what comes out is Bernoulli's equation.

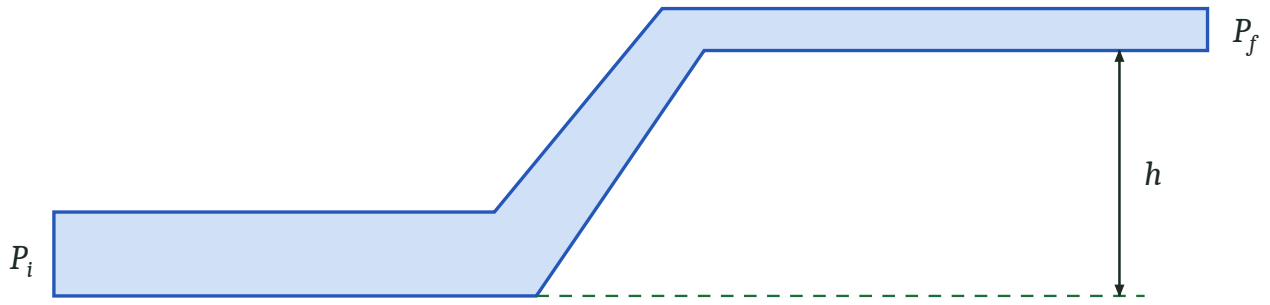
7. (A) Write the work–energy theorem for the lump as $\tfrac{1}{2}mv_f^2 - \tfrac{1}{2}mv_i^2 = -[\,\Delta\text{PE} + \Delta P \cdot V\,]$, and say which work the last term is. (B) Divide every term by V and say what m/V becomes. (C) Collect the i's on one side and the f's on the other, and write out Bernoulli's equation.

8. Name each of the three terms, and say what quantity each one is per unit of what. Which of the three ran the whole of Note 1, and what has to be true of the flow for the other two to drop out?

9. A horizontal pipe narrows. The water moves at 2.0 m/s in the wide part, where the pressure is 150 kPa, and the narrow part has **half** the area. Take $\rho = 1000$ kg/m³ and find the pressure in the narrow part.

When the pipe also rises

So far the pipe has been flat and the ρgh terms have cancelled. Let them not cancel. Here the pipe **narrows** and **climbs** at the same time, so the fluid gains kinetic energy *and* gains potential energy — and both bills land on the same account. Grouped the way it is easiest to read the signs off, Bernoulli says $\mathbf{\Delta KE = -[\,\Delta PE + \Delta \mathit{P}\,]}$, all three per unit volume.

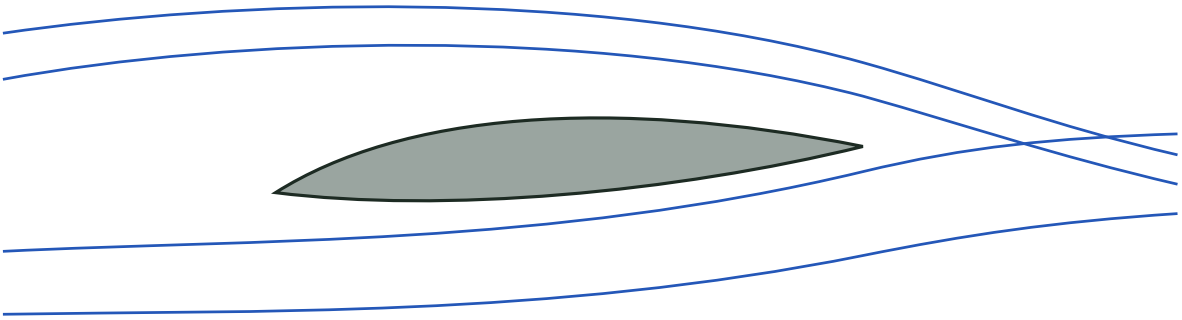


The narrow upper section is half the area of the lower one. Draw the speed arrow in each, in the right ratio, and mark beside each one whether the pressure there is high or low. The dashed line is the datum *h* is measured from; only the difference in height matters.

10. For the pipe in the figure, which narrows and climbs: (A) what is the sign of ΔKE, and why? (B) what is the sign of ΔPE? (C) what do those two force the sign of Δ*P* to be, and roughly how big?
11. Compare it with the same pipe kept flat. Is the pressure drop larger or smaller here, and which term is responsible for the difference?
12. Name one everyday consequence of this, and say which of the three Bernoulli terms causes it.

The wing

A wing is the famous case, and it is worth getting right rather than getting slick. The air passing over the upper surface moves **faster** than the air passing underneath; by Bernoulli, faster air is at **lower pressure**; so the pressure underneath exceeds the pressure above, and that difference, acting over the whole area of the wing, is **lift**. That much is right. The usual explanation of *why* the upper air is faster is not.



The streamlines are drawn crowded over the top and spread out underneath. Mark which flow is faster and which is slower, and which side is therefore at the lower pressure. Then draw the lift, and the arrow for what the wing does to the air at the trailing edge.

13. State the Bernoulli account of lift in two sentences: what is faster, what is therefore lower, and what the difference acts over.

14. "The air over the top has further to go and has to meet the air from underneath at the trailing edge." Is that right? Give **two** separate pieces of evidence for your answer.

15. If that is not the reason, what *does* set up the speed difference? Say how the Newton account and the Bernoulli account are related.

FOR PHY 151 — WITH CALCULUS

Bernoulli as an integral along a streamline. For steady flow of an inviscid, incompressible fluid, Newton's second law applied to a fluid element along its own streamline is Euler's equation, $\rho v\, dv/ds = -dP/ds - \rho g\, dh/ds$, with s measuring distance along the streamline. Multiply through by ds and integrate from one point to another: every term is an exact differential, so the integral collapses to $\tfrac{1}{2}\rho v^2 + P + \rho g h = \text{constant}$. Written this way the assumptions are visible rather than buried — **steady** (no $\partial v/\partial t$ term), **inviscid** (no friction term to integrate), **incompressible** (ρ comes out of the integral), and **along one streamline** (the constant may differ from streamline to streamline). Every failure of Bernoulli you will meet is one of those four being false.

16. A tank of water has a small hole a depth d below the surface. Apply Bernoulli from the surface to the hole, argue that both pressures are atmospheric and that the surface speed is negligible, and obtain **Torricelli's result**.

17. That is the same speed as an object dropped from height d. Say why, in terms of which Bernoulli terms survived.

18. The tank drains, so d falls. Write dd/dt in terms of the hole area a, the tank area A and d, and say what shape of curve $d(t)$ is.

Where this sits. Continuity and Bernoulli close Module 6. The method of Note 1 — weighing a sample twice — is what you take into the **Buoyancy Lab**; the pressure-versus-speed result from this note is what makes the flow demonstrations in class work the way they do.

BACK TO YOUR PREDICTION

The pressure in the narrow section is **lower**. Continuity forces the water to speed up there, and the extra kinetic energy is paid for out of pressure. If you predicted "higher" you are in good company — the squeeze feels like it should pressurise the water — but the standpipe over the throat is the short one, and the pressure comes back up again once the pipe widens and the water slows.

Check yourself

1. Water flows at 3.0 m/s through a pipe of cross-sectional area 0.020 m². The pipe narrows to 0.0050 m². What is the speed in the narrow section, in m/s?

2. In that narrow section, compared with the wide section, the pressure is

i. higher, because the same water is forced through a smaller opening

ii. lower, because the water is moving faster there

iii. the same, because the pipe is horizontal

iv. impossible to say without the density

3. Which term of Bernoulli's equation is the one that carries over unchanged from the fluid statics of Note 1?

i. $\frac{1}{2}\rho v^2$

ii. $\rho g h$

iii. P

iv. none of them — statics is a separate result

4. A pipe narrows *and* rises. Taking $\Delta KE = -[\Delta PE + \Delta P]$, the pressure change ΔP is

i. positive, and small

ii. negative, and smaller in size than for the same pipe kept flat

iii. negative, and larger in size than for the same pipe kept flat

iv. zero, because the two gains cancel

Your notes

BEFORE THE VIDEO

Guided notes — blank (PDF)

print or open on a tablet; record your prediction, then draw the figures and work the problems as the video goes through them.

AFTER THE MODULE

Completed notes (PDF)

released when Module 6 closes.

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